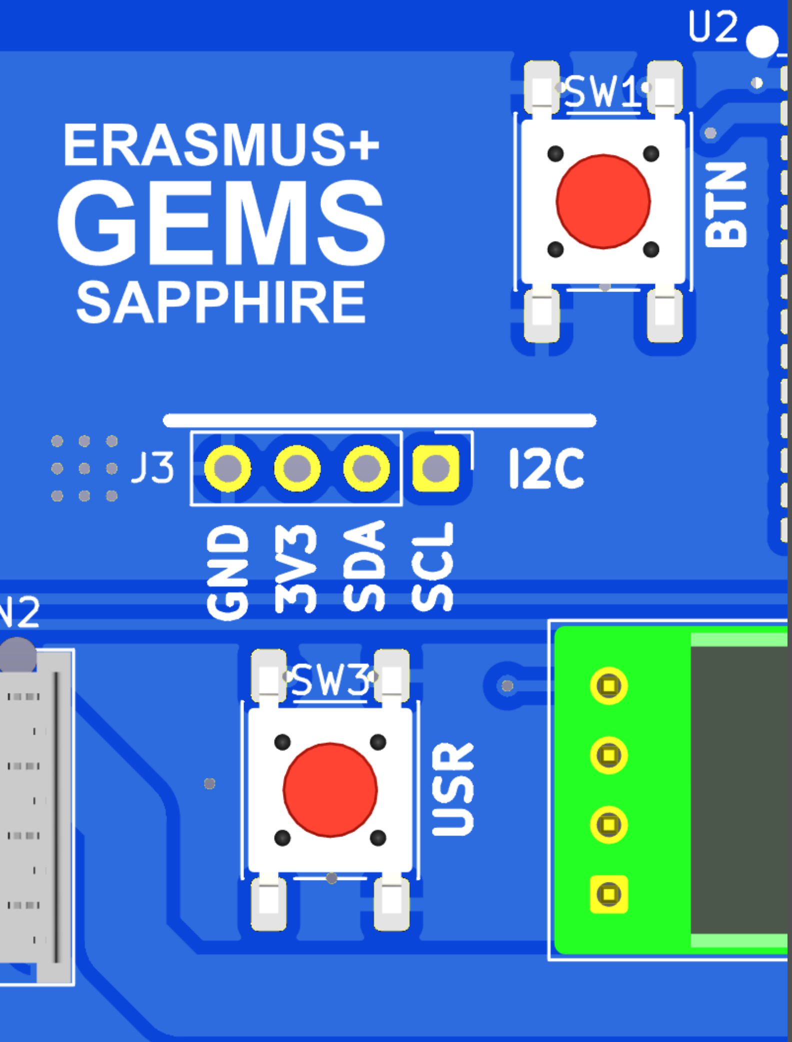

Communication Hardware Design

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Learing Objectives

1. **PLCs (Programmable Logic Controllers) and communication interfaces**
2. HMI (Human-Machine Interface) devices
3. Industrial networking hardware
4. Best Practices on PCB design techniques for industrial communication hardware

PLCs (Programmable Logic Controllers) and communication interfaces

Programmable Logic Controller (PLC):

- A PLC is an industrial-grade computer designed to control and automate machinery and processes.
- It is programmable, meaning users can write and upload custom programs to control specific tasks and operations.
- PLCs are highly reliable and are used for tasks ranging from simple machine control to complex process automation

PLCs (Programmable Logic Controllers) and communication interfaces

PLC as communication enabler:

Digital Inputs/Outputs (I/O): PLCs receive signals from digital sensors (like switches) and control digital actuators (like relays or solenoids). Digital I/O interfaces handle on/off signals.

Analog Inputs/Outputs (I/O): PLCs can also process analog signals (like temperature or pressure readings) and control devices that require variable signals (like variable-speed drives).

Serial Communication: PLCs often use RS-232, RS-485, or RS-422 serial communication protocols for connecting to other devices and systems.

Fieldbus Networks: Protocols like Profibus & Modbus, are used to connect PLCs with various field devices and sensors.

Industrial Ethernet: Modern PLCs often use Ethernet-based protocols such as EtherNet/IP, Profinet, or OPC UA to connect with other PLCs, computers, and networked devices.

HMI (Human-Machine Interface) devices and communication protocols

Human-Machine Interfaces (HMIs) are crucial components in industrial automation that enable communication between operators and PLCs or other control systems.

HMIs provide a graphical interface for users to interact with machinery and processes, monitor system status, and make adjustments.

HMI (Human-Machine Interface) devices and communication protocols

Graphical User Interface (GUI)

- **Visual Display:** HMIs present information visually using screens, often equipped with touch panels. This includes real-time data, process graphs, system alarms, and control options.
- **User Interaction**: Operators can interact with the system through buttons, sliders, and other controls on the HMI screen. This interaction allows them to start/stop processes, adjust setpoints, and respond to alarms.

HMI (Human-Machine Interface) devices and communication protocols

Real-Time Monitoring

- **Status Updates:** HMIs continuously receive and display real-time data from the PLC, such as machine status, operational parameters, and sensor readings
- **Alerts and Alarms:** HMIs can display alerts and alarms generated by the PLC, helping operators respond quickly to potential issues or abnormal conditions

Industrial networking hardware

Industrial networking hardware is a crucial component in industrial environments, enabling the collection, monitoring, and control of data from various processes and equipment

Industrial networking hardware

Industrial Networking Hardware facilitates communication and data exchange between devices and systems within an industrial environment, ensuring that processes are coordinated and data is shared effectively.

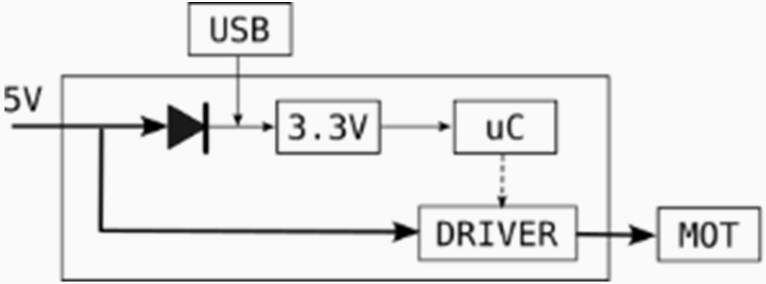
Components:

- Industrial Ethernet Switches
- Routers and Gateways
- Fieldbus Systems
- Wireless Network Devices
- Network Interface Cards (NICs)
- IoT Devices

Best Practices on PCB design techniques for industrial communication hardware

Requirements Definition

- Electrical Requirements: Voltage, current, signal frequencies, and power requirements.
- Mechanical Constraints: Size, shape, mounting holes, and enclosure considerations.
- Thermal Management: Heat dissipation needs based on power consumption.



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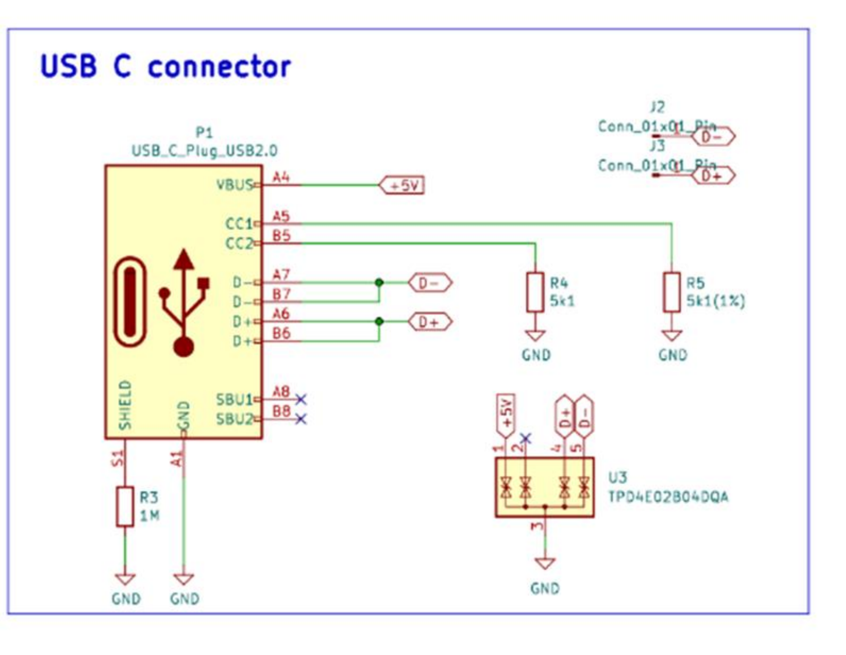
Schematic Design

- Component Selection: Choose components with suitable specifications for your design.
- Circuit Design: Create a schematic diagram, ensuring all connections are correct and logical.
- Component Placement: Place components with consideration for signal flow and ease of routing



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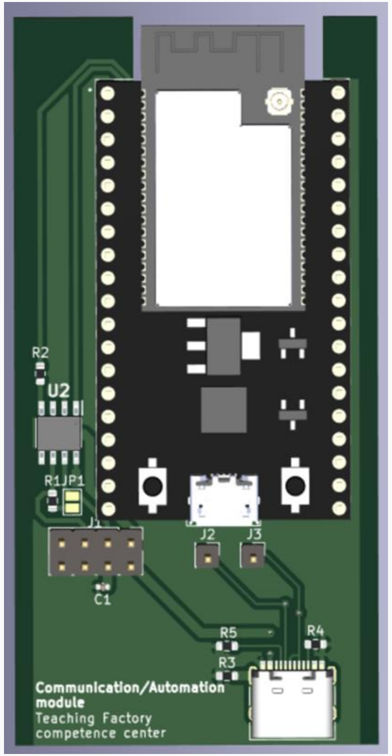
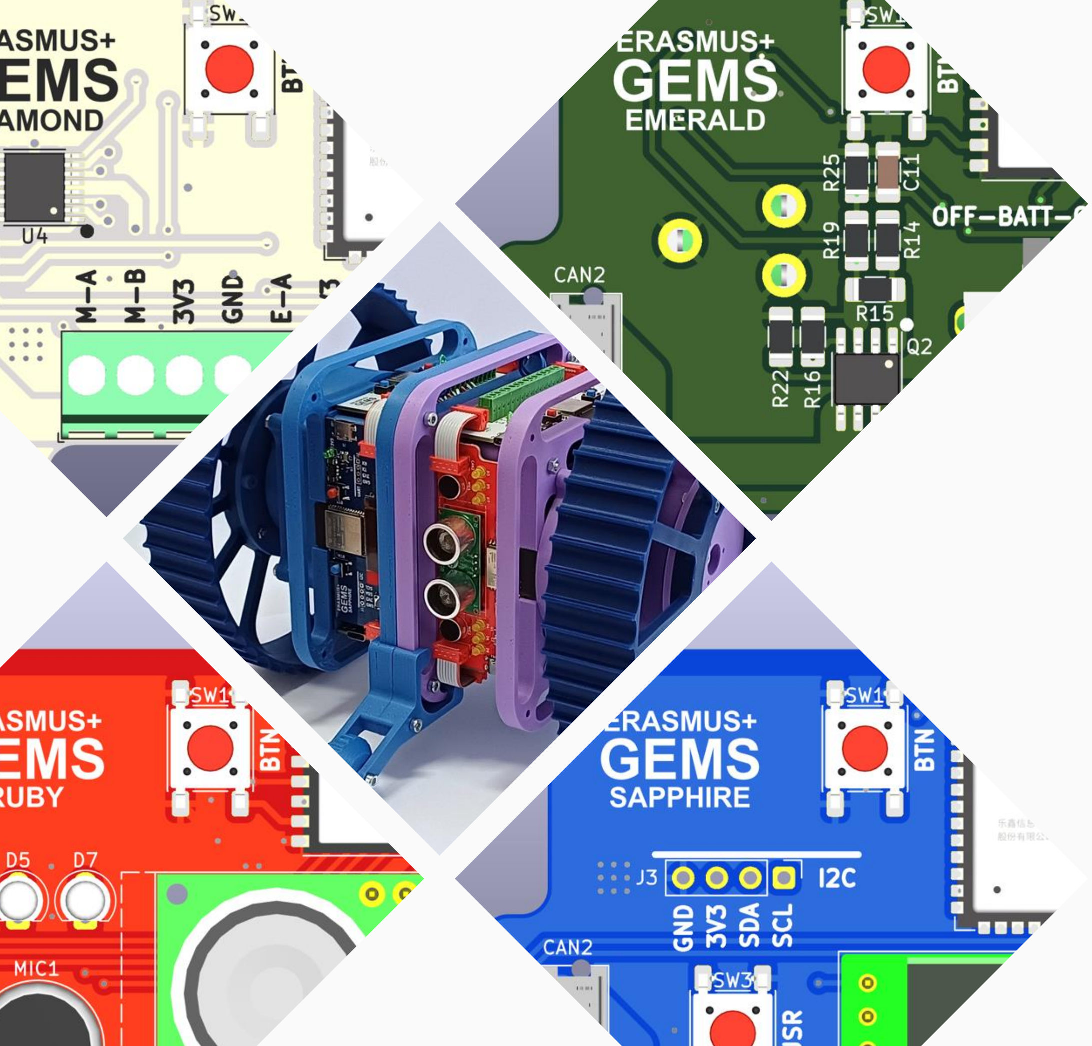
PCB Layout

- Layer Stack-Up: Decide on the number of layers based on complexity and signal integrity requirements.
- Common stack-ups include single-layer, double-layer, and multi-layer designs.
- Component Placement: Position components to minimize trace lengths, reduce noise, and simplify routing.Signal Traces: Keep traces short and direct





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Thank you for watching!

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